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SQL

Power BI

Excel

Python

Pizza Sales Data Analysis

Leveraging SQL to transform raw point-of-sale data into actionable business intelligence.

Map temporal sales trends and daily operational demands.

Identify best- and worst-performing products by volume and revenue.

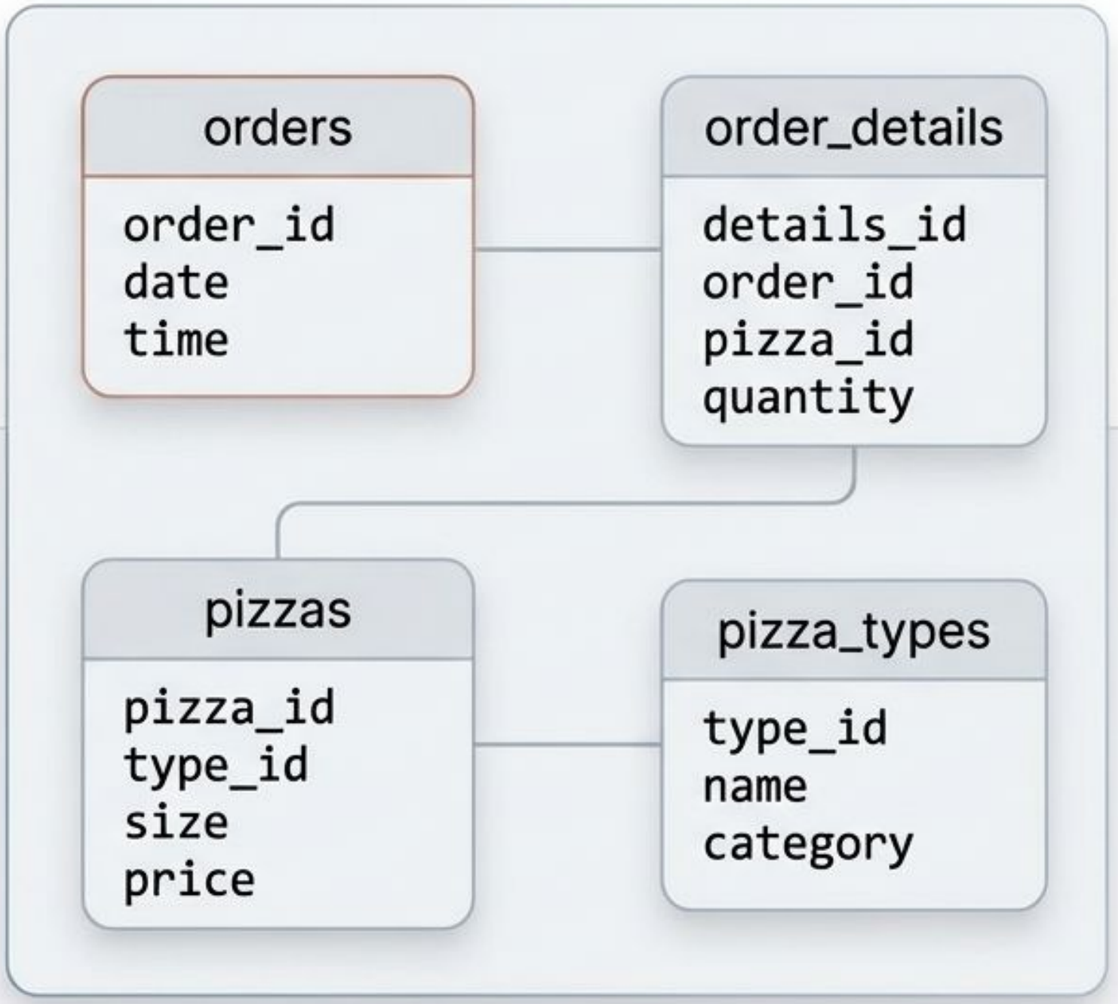
Uncover deep revenue drivers and category contributions.

Data Architecture & Schema setup

```sql
create database pizzahut;

create table orders (
order_id int not null,
order_date date not null,
order_time time not null,
primary key(order_id)
);

create table order_details (
order_details_id int not null,
order_id int not null,
pizza_id text not null,
quantity int not null,
primary key(order_details_id)
);
```



Core KPIs: Establishing Baseline Volume and Value

Volume

21,350 Total Orders

```
select count(order_id) as
total_orders
from orders;
```

```
| total_orders |
+--------------+
|        21350 |
+--------------+
```

Value

$817,860 Total Revenue

```
select round(sum(pizzas.price *
order_details.quantity),2) as
total_revenue
from order_details
join pizzas
  on order_details.pizza_id =
pizzas.pizza_id;
```

```
| total_revenue |
+---------------+
|     817860.05 |
+---------------+
```

The business successfully processed over 21,000 individual orders, generating nearly $818k in topline revenue during the analyzed period.

Product Preferences: Premium Pricing and Dominant Sizes

Highest-Priced Pizza

```sql
-- Identify the highest-priced pizza.

select pizza_types.name, pizzas.price from
pizza_types join pizzas on
pizza_types.pizza_type_id = pizzas.pizza_type_id order by
pizzas.price desc limit 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

name	price
The Gresk Pizza	35.95

The Greek Pizza holds the top price point at $35.95.

Most Common Pizza Size

```sql
-- Identify the most common pizza size ordered.

select pizzas.size, count(order_details.order_details_id) as order_count from
pizzas join order_details on
pizzas.pizza_id = order_details.pizza_id group by
pizzas.size order by order_count desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

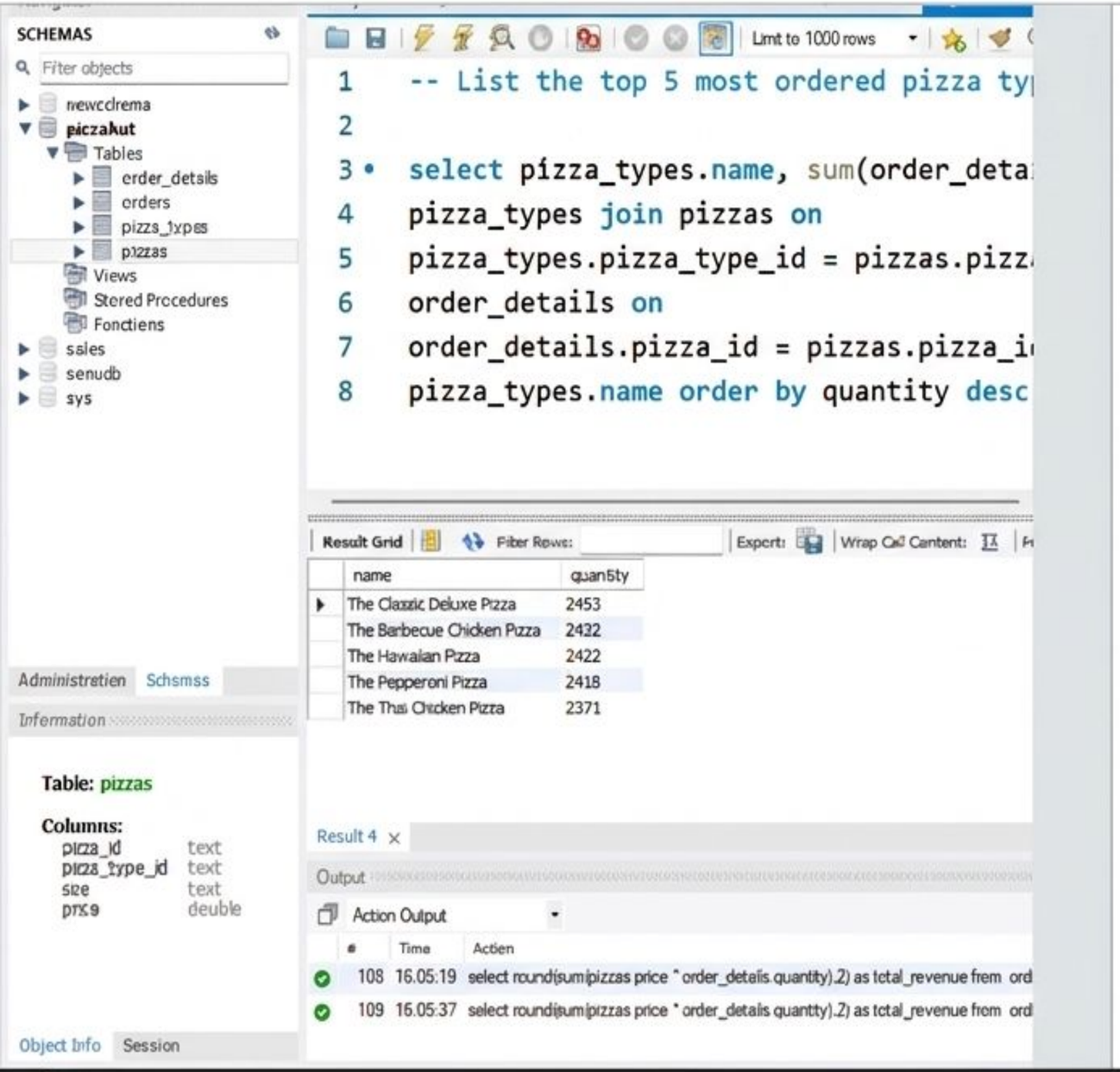
size	order_count
L	18356
M	16325
S	11137
XL	544
XXL	28

Large (L) is overwhelmingly the most popular size with 18,526 units sold.

Customer preference skews heavily toward the "Large" format, which makes up the vast majority of all volume.

Inventory management should prioritize Large-sized ingredients and boxes.

Top Sellers by Pure Order Volume



1 The Classic Deluxe Pizza (2,453)

2 The Barbecue Chicken Pizza (2,432)

3 The Hawaiian Pizza (2,422)

The 'Classic Deluxe' and 'Barbecue Chicken' pizzas lead the menu in raw sales volume. The top three are separated by less than 40 units, indicating strong, distributed demand across familiar profiles.

Category Dynamics: Menu Offerings vs. Actual Volume

Menu Offerings

```sql
-- Join relevant tables to find the category-wise distribution of pizzas.

select category, count(name) from
pizza_types group by
category;
```

category	count(name)
Chicken	6
Classic	8
Supreme	9
Veggie	9

Supreme (9)

Veggie (9)

Classic (8)

Chicken (6)

Sales Volume

```sql
-- Join the necessary tables to find the total quantity
-- of each pizza category ordered.

select pizza_types.category, sum(order_details.quantity) as quantity from
pizza_types join pizzas on
pizza_types.pizza_type_id = pizzas.pizza_type_id join
order_details on
order_details.pizza_id = pizzas.pizza_id group by
pizza_types.category order by quantity desc;
```

category	quantity
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

Classic (14,888)

Supreme (11,987)

Veggie (11,649)

Chicken (11,050)

A fascinating operational mismatch: While 'Supreme' and 'Veggie' offer the most menu variety, the 'Classic' category dominates total volume. Customers prefer familiar classics despite having fewer options to choose from.

Temporal Demand: Mapping the Daily Rushes

MySQL Workbench

SQL editor

```
1   -- Determine the distribution of orders by hour of the day.
2   select hour(order_time) as hour,
3         count(order_id) order_count
4   from orders
5   group by hour(order_time);
6
```

Result Grid

hour	order_count
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336
18	2399
19	2009
20	1642

Clear twin-peak demand. Staffing and kitchen prep must be heavily optimized for the 12-1 PM and 5-7 PM windows, while mid-afternoon lulls offer time for restocking.

Daily Operational Load (Subquery Aggregation)

```
-- Group the orders by date and calculate the average number of
pizzas ordered per day.
select round(avg(quantity_sum),0) as avrg_pizza_ordered_per_day
  from
    (select orders.order_date,
       sum(order_details.quantity) as quantity_sum
     from orders
     join order_details
       on orders.order_id = order_details.order_id
    group by orders.order_date) as order_qty;
```

Utilized a nested subquery to first calculate individual daily totals, then applied an outer aggregation to find the baseline average.

138

Average Pizzas Per Day

On average, the kitchen needs to be prepped to produce 138 pizzas daily. This forms the baseline metric for daily dough and ingredient preparation.

Revenue Drivers: Shifting from Volume to Value

SQL Query & Output

```sql
-- Determine the top 3 most ordered pizza types based on rev
select pizza_types.name, sum(order_details.quantity * pizzas
pizza_types join pizzas on
pizza_types.pizza_type_id = pizzas.pizza_type_id join
order_details on
pizzas.pizza_id = order_details.pizza_id group by
pizza_types.name order by revenue desc limit 3;
```

Result Grid | Filter Rows: | Experts | Wrap Call Content: | Fetch rews:

name	revsnue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42758
The California Chicken Pizza	41409.5

The Financial Reality

1. Thai Chicken — $43,434

2. Barbecue Chicken — $42,768

3. California Chicken — $41,409

Despite 'Classic Deluxe' leading in raw unit volume, the 'Thai Chicken' pizza claims the #1 spot for gross revenue. High-value specialty pizzas are the true financial engines of the menu.

Category Revenue Contribution (Advanced Subqueries)

SQL Query & Calculation

MySQL Workbench

```sql
-- Calculate the percentage contribution of each pizza type to total revenue.
select pizza_types.category,
    round((sum(order_details.quantity * pizzas.price) /
        (select round(sum(pizzas.price * order_details.quantity),2) as total
          from order_details
          join pizzas on order_details.pizza_id = pizzas.pizza_id))
 * 100, 2) as revenue_percentage
from pizza_types
join pizzas
  on pizza_types.pizza_type_id = pizzas.pizza
join order_details
  on pizzas.pizza_id = order_details.pizza_id
group by pizza_types.category
order by revenue_percentage desc;
```

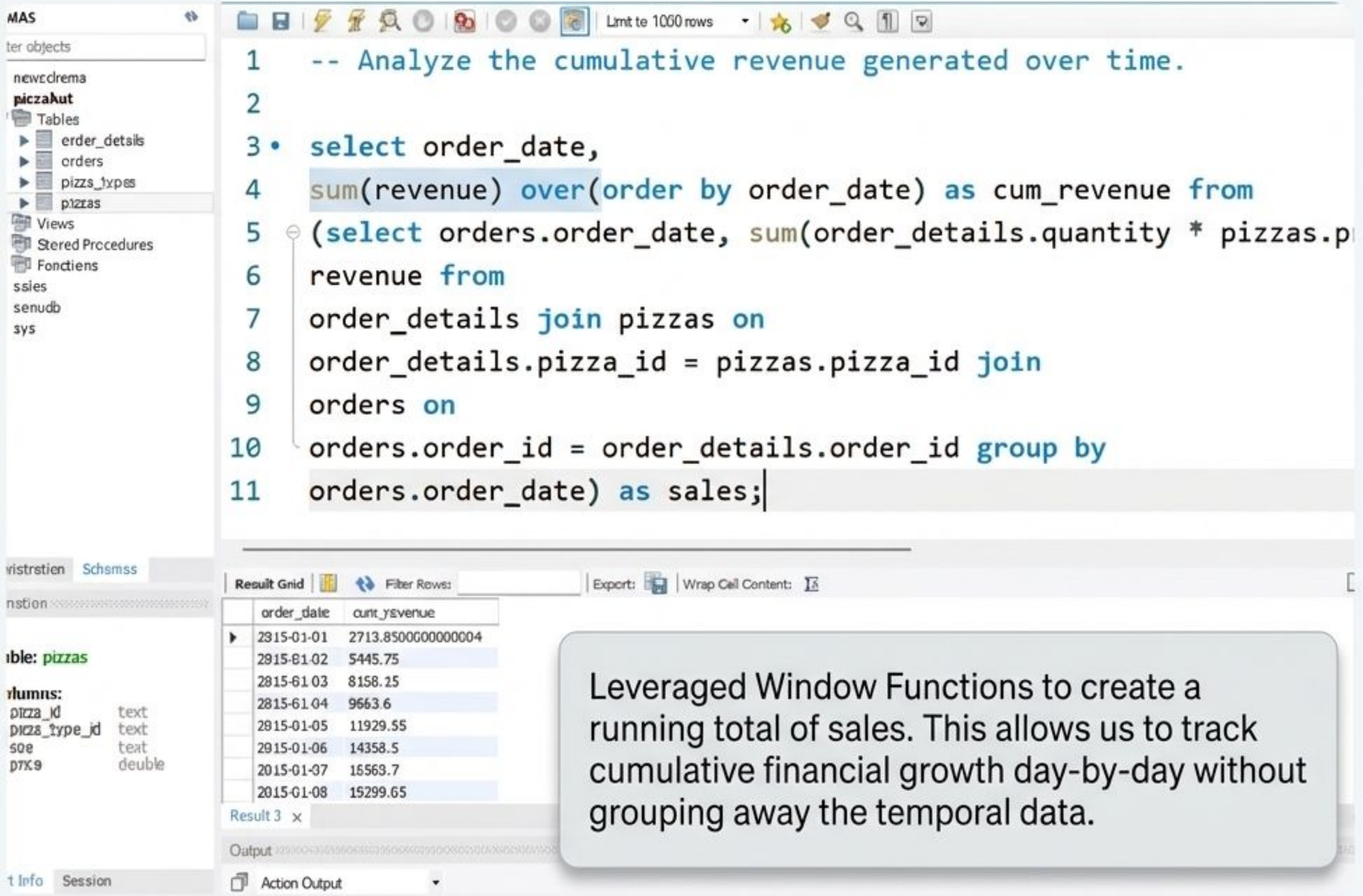
Complex subquery serves as the total revenue denominator for percentage calculation.

Utilized a complex subquery within the main query to determine the overall total revenue, which then serves as the denominator for calculating each category's percentage contribution.

Category Balance

An incredibly balanced menu. No single category monopolizes revenue; all four sit comfortably between 23% and 27%. This indicates a healthy, diversified product catalog that minimizes reliance on a single demographic.

Tracking Cumulative Growth with Window Functions



Leveraged Window Functions to create a running total of sales. This allows us to track cumulative financial growth day-by-day without grouping away the temporal data.

By January 8th alone, the business had **cumulatively generated over $19,399**, demonstrating strong, consistent daily cash flow at the start of the year.

Isolating Category Champions (CTEs & Ranking)

```
-- Determine the top 3 most ordered pizza types based on revenue for each pizza category.
WITH ( @pizza_category) {
SELECT category, top, category
    JOIN category in pizzaalona_=> pizza.specific
    JOIN category = in.revenue
    GROUP BY category, revenue
} CTE  ( revenue) {
  WHERE category = rank() over(partition by category order by revenue desc)
}
FROM top, category ;
```

Employed a Common Table Expression (CTE) combined with a Ranking Window Function to partition the data by category, cleanly isolating the top 3 revenue performers within each distinct menu segment.

This query empowers regional managers to identify the ‘anchor’ products for **localized marketing campaigns** within any specific category.

Executive Insights & Operational Strategy

Product & Marketing Strategy

- Focus inventory heavily on Large sizes.
- Heavily market the Thai Chicken (top revenue) and Classic Deluxe (top volume) as the menu anchors.

Operations & Staffing

- Optimize kitchen staffing exclusively around the twin peaks: 12-1 PM and 5-7 PM.
- Set daily ingredient prep baselines to accommodate an average of 138 pizzas.

Menu Optimization

- Maintain current category ratios. The revenue split (23-27% per category) shows perfect balance.
- Consider trimming unique options in Supreme/Veggie since Classic drives more volume with fewer options.

Thank You.

Always exploring data, currently open for new analytical opportunities and constructive feedback.

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View the full repository and raw queries online.